Namal University, Mianwali

Department of Computer Science

Project Proposal for

University Timetable Management System

CSC-271: Database Systems

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Submitted By:

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1. Introduction

In any education system, time management is very important for both teachers and students. A managed schedule helps everyone stay organized and focused. Therefore, managing university timetables manually through spreadsheets and emails is a time consuming and error-prone process. Moreover, changing a class becomes a problem with manual process. The University Timetable Management System is a web-based application that solves these problems by centralizing timetable creation and management for university. It allows administrators to create and publish timetables, faculty to view their assigned schedules and submit schedule change requests, and students to access their class timetables all from a single platform.

2. Problem Statement

The current manual timetable process suffers from the following issues:

- Venue conflicts: The same classroom is accidentally assigned to two courses at the same time.
- Instructor conflicts: A faculty member is scheduled for two courses simultaneously.
- No central system: Timetable data is scattered with no single source.
- No Change Request Mechanism: Faculty have no formal way to request changes to their assigned schedule, forcing them to make use of emails or verbal communication.

These problems lead to schedule mishaps, wastage of administrative time. To overcome these gaps, a database based system with role based access and constraints should be implemented.

3. Objectives

The key objectives of the University Timetable Management System are [1]:

1. Design a normalized relational database schema for all entities.
2. Implement venue and instructor conflict detection.
3. Enforce Role-Based Access Control for Admin, Faculty, and Student users.
4. Enable administrators to create and publish timetables.
5. Allow faculty and students to view their respective schedules.
6. Enable faculty to submit schedule change requests for administrator review.

4. Scope

The following features are included in the system [1]:

- **Timetable Management:** Create, edit, delete, and publish timetables and export to CSV.
- **Conflict Detection:** Venue double-booking and instructor overlap detection.
- **User Management:** Registration, login, RBAC, profile management, user deactivation.

- **Resource Management:** Venue and lab database with conflict prevention.
- **Course Management:** Course registry linked to departments and instructors.
- **Dashboards:** Role-specific views for Admin, Faculty, and Student.

The following are explicitly excluded:

- Automatic timetable generation.
- Timetable archiving and versioning.
- All reporting (summary, utilization, audit log).
- Soft-delete, MFA, bulk user import, timetable import, scheduled/in-app notifications, conflict resolution suggestions.

5. Data Collection

Data required to test the system will be collected as follows:

- User data (faculty and students): obtained from kaggle/mendeley data and github.
- Course and department information: collected from university's website.
- Venue and lab details: manually created.
- Academic calendar and time slot configuration: set up by the Admin based on the university schedule.
- Sample/test data: synthetic data generated using Python(Faker).

6. Technology Stack

The following technologies have been selected based on their suitability for a relational database-driven web application:

Layer	**Technology**	**Purpose**
Database	MySQL 8.4+	Core relational database for storing and managing all data
Backend	Node.js + Express.js	Handles server-side logic, database queries, and communication between frontend and database
Frontend	React	User interface for students, faculty, and admins
Authentication	bcrypt / Argon2	Ensures passwords are stored securely using strong hashing algorithms
Version Control	Git / GitHub	Source code version control and collaborative development

Table 1: Technology Stack

7. System Functionality

The following functionalities describe the core operations the system will support, based on the defined scope [1].

1. Administrators can create, edit, delete, and publish timetables by assigning courses, instructors, venues, and time slots.
2. The system detects and does not let the administrator select a venue or instructor at the same time slot.
3. Administrators can manage user accounts including creation, deactivation, and role assignment.
4. Users log in with email and password and their access is restricted based on their role.
5. Faculty can view only their assigned courses and schedule.
6. Faculty can submit a schedule change request specifying the request type, preferred time slot, venue, and reason.
7. Students can view only their enrolled class timetable.
8. Administrators can manage venues, labs and courses.
9. Administrators can review pending schedule change requests and approve or reject them.
10. Timetable data can be exported to CSV.

8. User Classes

The system defines three user roles, each with distinct access levels within the system.

Role	Description	Access Level
Admin	Manages the entire system	Full access: timetables, users, resources, settings
Faculty	Teaching staff	View own assigned schedule only, submit schedule change requests
Student	Enrolled students	View own class timetable only

Table 2: User Classes

9. Task Division

The project has been divided among team members keeping balanced contribution and responsibilities. All members will also be collaborating with each other throughout the project helping each other with their task.

Phase	**Hammad Shabbir**	**Ahmer Sultan**
Proposal Report	Introduction, Problem Statement, Objectives, Scope	Data Collection, Technology Stack, Functionality, User Classes Conclusion
Database Design	ER Diagram, Users, Courses, Departments Tables	ER Diagram, Timetable, Venues, Time Slots Tables
Backend	Authentication, RBAC, User Management	Timetable Management, Conflict Detection, Schedule Change Requests
Frontend	Admin Panel, User Management Pages	Faculty & Student Dashboards Pages

Table 3: Task Division

10. Conclusion

The University Timetable Management System will replace the existing manual and unstable scheduling process with an organized database-based system. The University Timetable Management System will decrease errors in scheduling and administration overhead due to enforcing role-based access, conflict detection, and centralized data management. The project scope is suitable as it relates to the course Database Systems because it offers practical implementation of concepts such as schema normalization, relational integrity, role-based access control and constraint enforcement.

References

[1] H. Shabbir and A. Sultan, "Software requirements specification: University timetable management system," Namal University, Mianwali, Pakistan, 2025.

[2] "University timetable management system," GitHub. Available: `https://github.com/hammad-013/university-timetable-management-system`

Appendix 1: AI Tools Used

Tool	Purpose	Prompts Used
Claude	Recommendation about the system	"This is the SRS of our software engineering project and we are continuing the same project in the Database Systems. The scope of project is bit reduced to make it feasible for the course. Automatic timetable generation, archiving, course coordinator user, has been excluded. Only student, faculty and admin are the users. Moreover, some other things can be excluded (recommend it to me). We need to make proposal for database systems proposal. Project should be feasible. So first recommend what do i need to keep in the system.", "This is the proposal, can you identify the problems and recommend the changes", "i cannot find data for now, what should i write in data collection section"
Grok	Structuring the document	"This table does not look good, change its look like table given below:", "where can i collect data from for database?"

Table 4: Tools Used in Proposal Development